Aarav Verma

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NOBE — Project Manager · Ghost Electric Motorcycles — Drivetrain Engineer · Tarform — Part-Time Mechanical Design Engineer Intern

CAD & HARDWARE PROJECTS



Full Drone + Computer System Feb 2026 – Present

NOBE — Engineering Lead

Designed a modular drone airframe in Fusion 360 with a Jetson housing and tool-free foldable leg motors on 4 snap-lock hinges. FEA under 12N thrust load confirmed FOS above 2.0 across a 50+ component BOM.

Fusion 360, FEA, Avionics Integration



Toroidal Propeller Design Feb 2026 – Present

NOBE — Engineering Lead

Redesigned a 2-blade propeller to a toroidal configuration in Fusion 360 — enclosing the blade-tip vortex path raised thrust efficiency ~11% at 4–6k RPM and cut acoustic signature below 60 dB at 1m hover, FOS above 2.0.

Fusion 360, FEA, Aerodynamics



Swingarm Assembly Aug 2025 – Jan 2026

Ghost Electric Motorcycles — Drivetrain Engineer

Reverse-engineered a rear suspension swingarm from the 2010 Kawasaki ZX-6R in SolidWorks to replicate OEM anti-squat ratio and pivot geometry within ±0.05 mm, then validated stiffness under braking, acceleration, and cornering loads.

SolidWorks, ANSYS FEA, MATLAB



GHOST — Drivetrain Frame Aug 2025 – Jan 2026

Ghost Electric Motorcycles — Drivetrain Engineer

Engineered a load-bearing drivetrain frame in SolidWorks to align motor, top rod, and swingarm to ±0.05 mm. ANSYS and MATLAB confirmed stiffness under peak torque, chain tension, and rider weight across all loading conditions.

SolidWorks, ANSYS FEA, MATLAB



Revolving Modular Hydroponic Tower 2024

UIUC Residence Hall Idea Fair — Designer & Builder

Designed a revolving modular hydroponic tower in Fusion 360, building the stacking system around FDM print tolerances for every interlocking joint. Printed and assembled the full prototype in-house.

Fusion 360, 3D Printing, DFM



Volant Demo Drone 2025 – Present

Volant (Personal Startup) — Founder & Designer

Designed a palm-sized, field-deployable drone airframe in Fusion 360 to FDM print tolerances as the hardware side of Volant, built to run in a 10-drone swarm tied into a fleet telemetry and mission dispatch platform.

Fusion 360, 3D Printing, UAV Design

SIMULATION & ANALYSIS — MATLAB / SIMULINK

Cascaded Flight Controller — Pitch / Roll / Yaw 2026

Simulink

Designed a cascaded rate/angle PID controller (Betaflight/PX4-style) in Simulink, deriving plant dynamics from first principles and tuning via manual and Ziegler–Nichols methods. Diagnosed and corrected a derivative-kick and noise-amplification problem at its root cause, then validated disturbance rejection against injected wind-gust and sensor-noise models.

Simulink, PID Control, MATLAB

Parametric Airfoil Structural Analysis 2026

MATLAB PDE Toolbox

Generated NACA 0012 airfoil geometry in MATLAB, modeled the solid in Fusion 360, and imported STEP geometry back into MATLAB for structural FEA under directional surface loads. Ran a mesh convergence study, caught a stress singularity at an unfilleted corner, and corrected it through CAD redesign to bring successive mesh results within 2% of each other.

MATLAB, FEA, Mesh Convergence, Fusion 360

Cantilever Bracket — Hand-Validated FEA 2026

MATLAB PDE Toolbox

Designed a hand-calc vs. FEA comparison experiment to validate tip deflection and bending stress predictions, catching a 10x error in the FEA load application by querying the true load-face area from the solved mesh. Reconciled corrected output to within 10% of hand calculations, attributing the remaining gap to bend-fillet compliance and hole stress concentration.

MATLAB, FEA, Mechanics of Materials

SOFTWARE PROJECTS

TVC Simulator github.com/Verma0118/TVC_Simulator_V1

Simulation Tool

Thrust Vector Control simulator modeling gimbal-actuated rocket engine dynamics for attitude control during powered flight.

Python, Simulation, Controls

SKILLS

CAD & Design: Siemens NX, SolidWorks, CATIA V5, Fusion 360, GD&T (ASME Y14.5), 3D Printing (FDM)

Simulation & Analysis: ANSYS FEA, MATLAB (PDE Toolbox), Simulink

Programming & Software: Python, TypeScript, React, C++, Git, ROS

Manufacturing: FDM 3D Printing, CNC Machining, Laser Cutting, Sheet Metal Fabrication

COURSEWORK — AE 140

Introduction to CAD for Aerospace Engineers. 10 modeled assemblies including a UAV design, spur gear, front sprocket, and multiple tutorial/homework exercises in Siemens NX.

Full interactive portfolio with rotatable 3D model viewers at verma0118.github.io